Assignment 1

Diabetes Risk Prediction using ML

1. Introduction

The objective of this project is to analyze a healthcare dataset and build predictive models to identify whether a patient is likely to have diabetes. Machine learning techniques such as **Logistic Regression** and **Neural Networks** are used to perform classification based on patient health attributes.

2. Dataset Description

The dataset was loaded as Pandas DataFrame for analysis. It consists of multiple health-related features such as:

- Age
- Body Mass Index (BMI)
- Exercise hours per week
- Other medical attributes
- Target variable: **Diabetes (0 = No, 1 = Yes)**

The dataset was found to be structured and suitable for machine learning tasks.

3. Data Preprocessing

3.1 Data Cleaning

- The patient_id column was removed as it does not contribute to prediction.
- Missing values and duplicates were checked and handled.
- The dataset was confirmed to be clean and consistent.

3.2 Feature Selection

- Input features (X) were separated from the target variable (y).
- The target variable is **diabetes status**.

3.3 Train-Test Split

- The dataset was split into:
    - **80% training data**
    - **20% testing data**

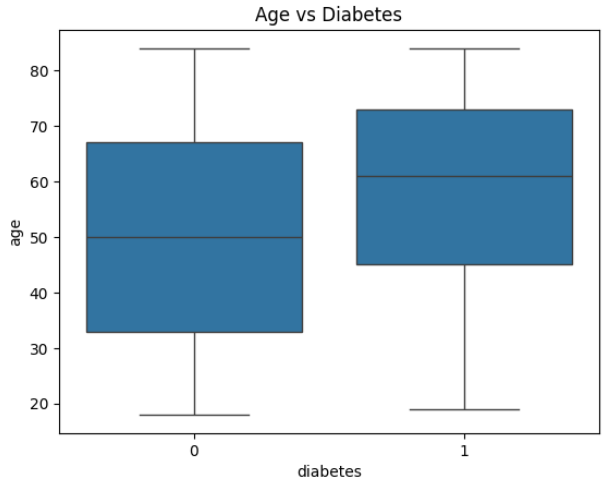
3.4 Feature Scaling

- Standardization was applied using StandardScaler to normalize the feature values.
- This improves model performance, especially for distance-based algorithms.

4. Exploratory Data Analysis (EDA)

4.1 Age vs Diabetes

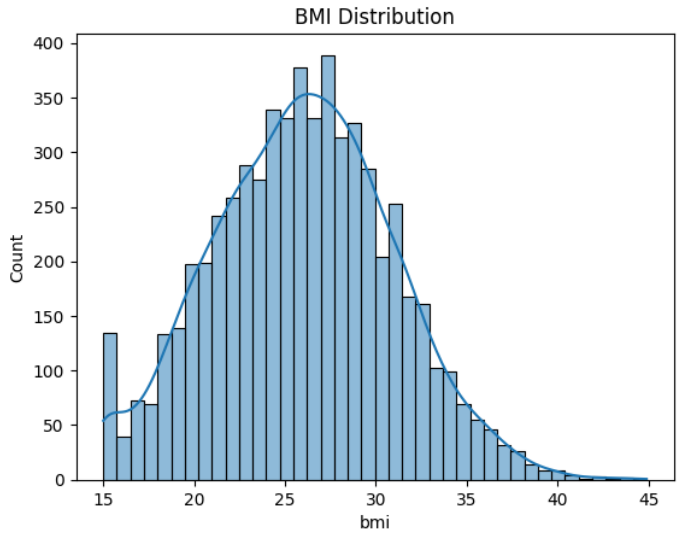
- A boxplot was used to compare age distribution between diabetic and non-diabetic patients.



Inference:
Older individuals show a higher likelihood of having diabetes, indicating age is a significant risk factor.

4.2 BMI Distribution

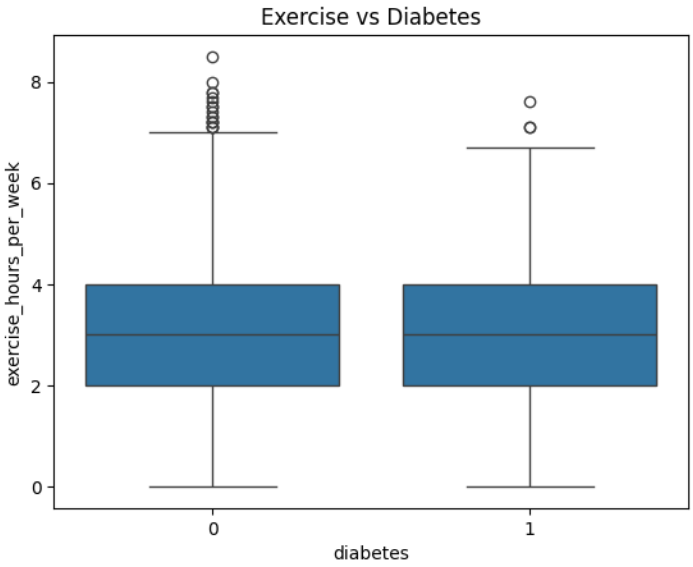
- A histogram was used to analyze BMI distribution.



Inference:
BMI shows a near normal distribution with slight skewness

4.3 Exercise vs Diabetes

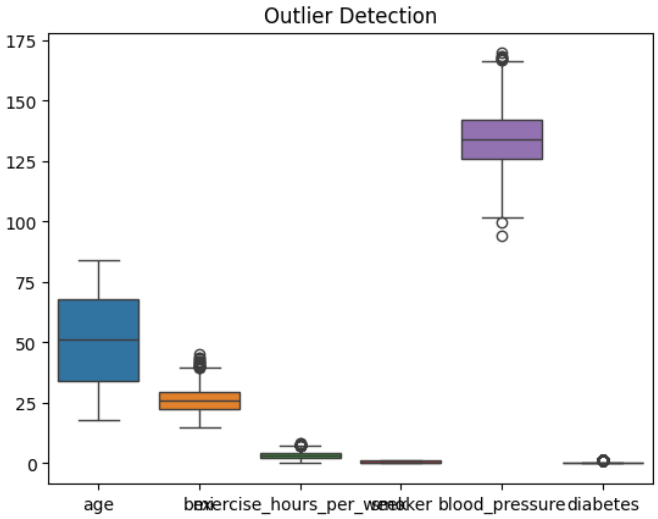
- A boxplot was used to compare exercise habits.



Inference:
Patients with lower physical activity levels are more likely to have diabetes, highlighting the importance of lifestyle.

4.4 Outlier Detection

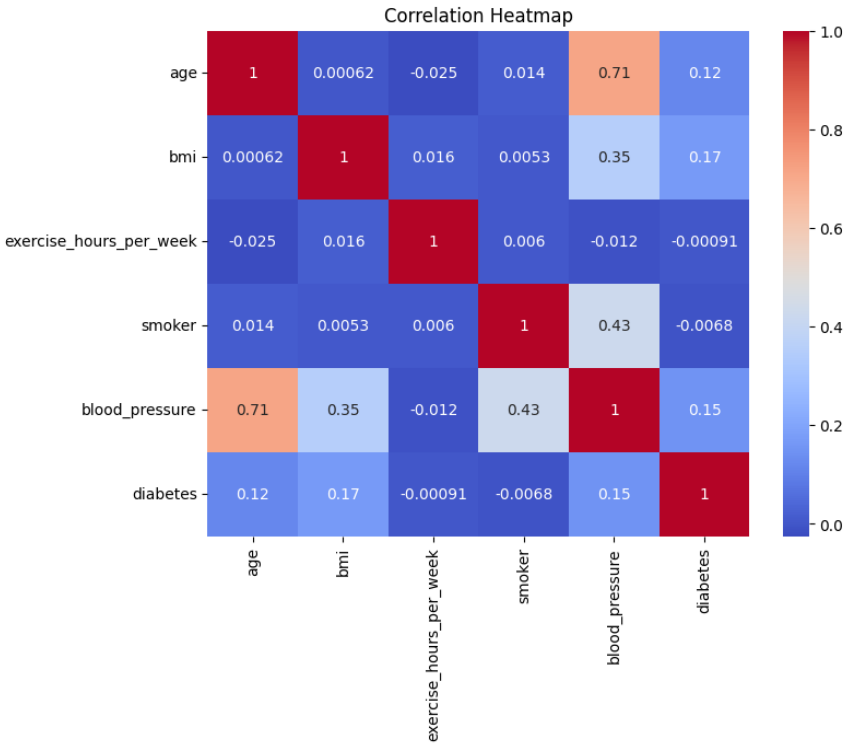
- A boxplot of all features was used to identify outliers.



Inference:
Some features contain outliers, which can cause model performance.

4.5 Correlation Analysis

- A heatmap was used to visualize correlations between features.



Inference:
Features such as BMI, age, and lifestyle factors show stronger correlation with diabetes.

5. Model Building

5.1 Logistic Regression

A **Logistic Regression** model was trained using the training dataset.

- The model learns relationships between input features and the target variable.
- Predictions were made on test data using:

y_pred_lr = model_lr.predict(X_test)

5.2 Neural Network

A neural network was built using TensorFlow/Keras:

- Input layer
- Hidden layers (16 and 8 neurons)
- Output layer (sigmoid activation)

The model was trained for 20 epochs to learn patterns in the dataset.

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6. Model Evaluation

6.1 Logistic Regression Performance

- Accuracy: ~90%
- Confusion matrix and classification report indicate strong performance.

Inference:
The model performs well in distinguishing between diabetic and non-diabetic patients.

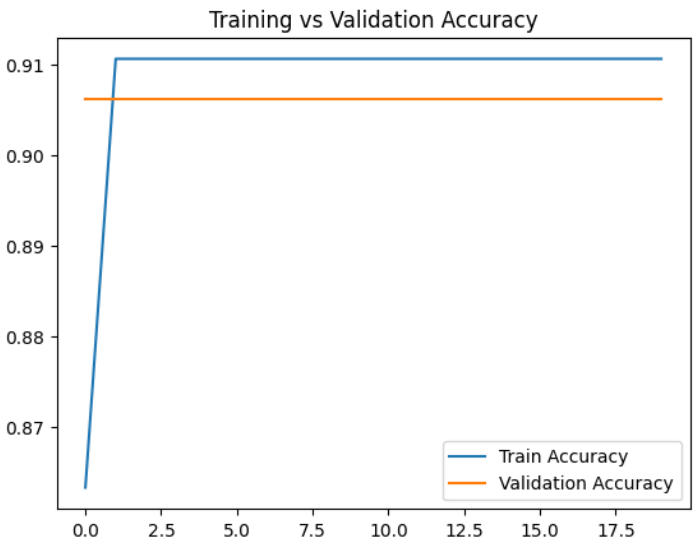
6.2 Neural Network Performance

- Accuracy: ~90%

Inference:
The neural network performs similarly to logistic regression, indicating that the dataset does not require a highly complex model.

6.3 Training vs Validation Accuracy

- Accuracy curves were plotted.



Inference:
Training and validation accuracy are closely aligned - minimal overfitting.

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7. Feature Importance

- Logistic regression coefficients were analyzed.

Inference:
Key influencing factors include:

- BMI
- Age
- Lifestyle factors (exercise)

These features significantly contribute to diabetes prediction.

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8. Comparison of Models

Model	Accuracy
Logistic Regression	~90%
Neural Network	~90%

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Logistic Regression Accuracy: 0.9091666666666667
Neural Network Accuracy: 0.909166693687439
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Inference:
Both models perform equally well. Logistic Regression is preferred due to its simplicity and interpretability.

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9. Conclusion

- The dataset was successfully analyzed and modeled.
- Age, BMI, and exercise levels are strong predictors of diabetes.
- Logistic Regression achieved high accuracy and proved to be efficient.
- Neural Networks did not provide significant improvement over simpler models.